CH 7 - Linear Diophantine Equations

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Recall the Extended Euclidean Algorithm

$$253x + 143y = d, d = \gcd(253, 143)$$

i	x	y	r	q
$i = 1$	1	0	253	0
$i = 2$	0	1	143	0
$i = 3$	1	−1	110	1
$i = 4$	−1	2	33	1
$i = 5$	4	−7	11	3
$i = 6$	−13	23	0	3

Diophantine Equations

💡 **Tip** — Simplest Linear Diophantine Equation: $ax = b$

📐 **Theorem — Linear Diophantine Equation Theorem, Part 1 (LDET 1)**
For all integers $a, b,$ and c, with a, b both not zero, the linear Diophantine equation

$$ax + by = c$$

(in variable x and y) has integer solution if and only if $d \mid c$, where $d = \gcd(a, b)$

Proof

Let $a, b, c \in \mathbb{Z}; a, b \neq 0; d = \gcd(a, b)$
We prove two implications:
1. $\implies$
   Suppose $\exists x_0, y_0 \in \mathbb{Z}, ax_0 + by_0 = c$

Since $d = \gcd(a, b)$, we have $d \mid a, d \mid b$.
Since $x_0, y_0 \in \mathbb{Z}$, by DIC, $d \mid (ax_0 + by_0)$

2. $\impliedby$ Suppose $d \mid c$.

Then by defintion $\exists l \in \mathbb{Z}$ s.t $c = l \cdot d$.
By Bézout's Lemma, $\exists s, t \in \mathbb{Z}$ s.t.
$as + bt = d$. Multiply the equation by $l \implies asl + btl = dl = a(ls) + b(lt) = c$.
Since $s, l, t \in \mathbb{Z}$, we have integer solution to the Diophantine equation, namely $x = ls, y = lt$

□

📐 **Theorem — Linear Diophantine Equation Theorem, Part 2 (LDET 2)**
Let a, b, c be integers with a, b both not zero, and define $d = \gcd(a, b)$. If $x = x_0$ and $y = y_0$ is one particular integer solution to the linear Diophantine equation $ax + by = c$, then the set of all solutions is given by

$$\left\{(x, y) : x = x_0 + \frac{b}{d}n, y = y_0 + \frac{a}{d}n, n \in \mathbb{Z}\right\}$$

Proof

Let $a, b, c \in \mathbb{Z}$ with $a, b \neq 0$. Let $d = \gcd(a, b)$
Suppose $x = x_0, y = y_0$ is one particular solution to LDE $ax + by = c$

Let $A = \left\{(x, y) : x = x_0 + \frac{b}{d}n, y = y_0 + \frac{a}{d}n, n \in \mathbb{Z}\right\}$
Let $B = \{(x, y) : ax + by = c, x, y \in \mathbb{Z}\}$

We want to show

1. $A \subseteq B$, suppose $(x, y) \in A$, then $x = x_0 + \frac{b}{d}n, y = y_0 + \frac{a}{d}n, n \in \mathbb{Z}$

Note, since $d \mid a, d \mid b \implies \frac{b}{d}, \frac{a}{d} \in \mathbb{Z}$

So $x = x_0 + \frac{b}{d}n \in \mathbb{Z}$ and $y = y_0 + \frac{a}{d}n \in \mathbb{Z}$

Now substitute in x, y to then LHS of the linear Diophantine equation.

Then $ax + by = a\left(x_0 + \frac{b}{d}n\right) + b\left(y_0 + \frac{a}{d}n\right) = ax_0 + \frac{ab}{d}n + by_0 - \frac{ab}{d}n$

$\implies ax_0 + by_0 = c.$

$\therefore (x, y) \in B \implies A \subseteq B$

2. $B \subseteq A$ consider $(x, y) \in B$, then $x, y \in \mathbb{Z}$ and $ax + by = c$.

We also have (x_0, y_0) is a solution to the LDE, so $ax_0 + by_0 = c$

Substract those equations: $ax + by - ax_0 - by_0 = 0 \implies a(x - x_0) + b(y - y_0) = c$

Then $a(x - x_0) = -b(y - y_0)$

Note, since $a, b \neq 0, d = \gcd(a, b) > 0, \frac{a}{d}$ and $-\frac{b}{d} \in \mathbb{Z}$

So $\frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0) \implies \frac{b}{d} \mid \left(\frac{a}{b}(x - x_0)\right)$

By Division by GCD, $\gcd\left(\frac{a}{d}, -\frac{b}{c}\right) = 1$

By Coprimeness and Divisibility, $\frac{b}{d} \mid (x - x_0)$.

By definition of divisibility, $\exists n \in \mathbb{Z}, x - x_0 = \frac{b}{d}n$ in other words, $x = x_0 + \frac{b}{d}n$.

Substitute $y - y_0 = \frac{b}{d}n$ and isolate: $-\frac{a}{d}\left(\frac{b}{d}n\right) = -\frac{b}{d}y - y_0 \implies y = y_0 - \frac{a}{d}n$

$\therefore (x, y) \in A$, so $B \subseteq A$

□

Examples:
Are there integer solutions to the following linear Diophantine equation:

1. $253x + 143y = 11$

ANS: YES $x = 4, y = -7$

2. $253x + 143y = 155$

ANS: LDET 1 says there exists a solution if and only if $11 \mid 155$.
However, $11 \nmid 155$. Hence there are no integer solutions

3. $253x + 143y = 154$

ANS: LDET 1 says there exists a solution if and only if $11 \mid 154$. $11 \mid (11 \cdot 14)$.
By multipling the equation of example 1 by 14:
$14 \cdot (253x + 143y) = 14 \cdot 11 = 253 \cdot (14x) + 143 \cdot (14y) = 154, x = 56, y = -98$

4. $343x + 259y = 658$

ANS: Has a solution, $x = -282, y = 376$

To find all solutions, we apply LEDT 2, the solution set is
$\{(x, y) : -282 + 37n, y = 376 - 49n, n \in \mathbb{Z}\}$

5. A customers has a large quantity of dimes and quarters. In how many ways can she pay exactly for an items that costs \$ 2.65?

ANS: Let x be number of quarters and y be number of dimes.

Consider LDE: $25x + 10y = 265$. We look for non-negative integer solutions.

By inspection, $x = 9, y = 4$ is one particular solution

By LDET 2, we have $\{x, y\} : 9 + 2n, 4 - 5n, n \geq 0$ We get $n = \{-4, -3, -2, -1, 0\}$ that satisfy the inequalities.